

ANGLIA RUSKIN UNIVERSITY

RESEARCH PROJECT PROPOSAL

Operationalizing Responsible Data Governance

A Comparative Documentary Study of Ethical Controls in Big-Data Business Decision-Making

Student	Koparthi Vinay Babu
Student ID (SID)	2454205
Module	MOD002695 – Research Methods
Module Leader	Jin Zhang
Main-text word count	2,885 words

P0-;[91. Introduction

1.1 Motivation and context

Big-data analytics is now part of the decision-making process for hiring credit, pricing, fraud, marketing and customer support. By merging huge and diverse data sets businesses can spot trends that would be difficult to see with traditional analysis. These capabilities can improve the quality of decisions and increase efficiency, but they also alter who is visible, how people are classified, and whether a choice can be understood or disputed. Poorly controlled analytics can leak personal information, perpetuate past discrimination, or hide accountability in a technically complicated system (Mittelstadt et al., 2016; Someh et al., 2019).

The research is driven by a practical shortcoming of current governance. Many organizations have statements of commitment to privacy, fairness and openness. But a statement of principle does not tell us whether risks are analyzed before deployment, monitored thereafter or corrected when harm happens. This is not just an ethical issue but also a commercial one: weak controls can result in regulatory action, reputational damage and loss of stakeholder trust. The suggested study therefore looks on operational evidence – impact assessment, bias testing, traceability, human review, audit and redress – rather than policy language as proof of responsible practice.

1.2 Objective

To assess the relationship between integrated data-governance measures and increased documented mitigation of privacy, fairness, transparency and accountability concerns in big-data business decision-making.

1.3 Objectives and expected results

1. Critically synthesize scholarly and regulatory research on ethical hazards and present governance practice.
2. Build an analytical framework for privacy assessment, fairness testing, transparency, human oversight and accountability audit.
3. Apply the framework to 12–20 organizational examples found through a transparent search and screening methodology.
4. Assess governance maturity against independently recorded evidence of mitigation, remediation or unaddressed harm.
5. Provide a reviewed evidence corpus, coding methodology, case-comparison matrix, hypothesis assessment and practical governance recommendations.

1.4 Research questions

1. What are the most common ethical risks associated with the organizational use of big-data analytics?
2. What governance controls are in place and how well are they implemented?
3. How is the level of governance maturity related to independently documented risk mitigation?
4. What combinations of controls seem most credible for responsible corporate practice?

1.5 Hypotheses and operational definitions

1. H1: instances with integrated governance controls will have more robust independently substantiated ethical-risk mitigation than instances depending primarily on compliance assertions.
2. H0: There will be no consistent difference in the documentation of mitigation between cases that incorporate controls and cases that rely primarily on compliance statements.

The integrated method is described by five control domains: privacy and data-protection assessment, fairness testing, openness or explainability, human oversight and remedy, and responsible audit. Each domain will be graded 0 (not present), 1 (stated or partial) or 2 (executed with supporting evidence) to give a Governance Maturity Score of 0–10. Mitigation will be scored separately: 0 = harm not resolved or no evidence of mitigation; 1 = planned, partial or self-reported remediation; 2 = remediation or ongoing monitoring confirmed by independent academic or regulatory source. H1 is rejected due to the separation of governance and outcome evidence.

2. Review of the Literature and Research Justification

2.1 Big data as an ethical and organizational system

While big data is often defined in terms of volume, velocity and diversity, these technical features do not explain how analytics delivers commercial benefit or harm. The results will be a function of data provenance, modelling choices, organizational incentives and the authority given to human decision makers. Hence, big-data analytics is conceptualized as a socio-technical phenomenon encompassing individuals, organizations and society (Someh et al., 2019). This perspective is crucial as an effective organizational decision may impose costs in terms of privacy, autonomy or discrimination on persons who neither designed nor manage the system.

2.2 Data stewardship, autonomy and privacy

Datasets acquired for one reason can be connected and inferred and reused in another context. Therefore, scale makes meaningful consent difficult and raises the risk of re-identification and function creep. Zwitter (2014) believes that the predictive capacity changes the ethical relation between data holders and data subjects. It is not only about secrecy, but also the power to classify and affect people without their awareness.

UK data-protection practice provides a useful baseline. The UK GDPR principles are lawfulness, fairness and openness, purpose limitation, data minimization, accuracy, storage limitation, security and accountability (Information Commissioner's Office, 2026). But legal compliance does not decide whether an inference is reasonable, or a corporate purpose is ethically justified. The research consequently supports the view that purpose controls, effect assessment and means of recourse should be considered alongside formal compliance.

2.3 Bias, fairness and accountability

Algorithmic judgements are selections of data, labels, objectives, thresholds, and tolerable error. Historical data may include prior inequities, and an acceptable mean performance may mask worse results for certain groups. As Mittelstadt et al. (2016) point out, it is difficult to assign responsibility due to opacity and inconclusive evidence. Martin (2019) goes a step further, claiming that algorithms are value-laden:

organizations are nonetheless responsible for design and deployment decisions even if a model is technically hard to grasp.

2.4 Current Practice: Principles to Controls

Today, legal obligations are mixed with voluntary norms and internal governance. The primary frameworks utilize distinct vocabulary but converge on lifetime risk management and provable accountability.

- *ICO / UK GDPR*
Key focus: Fair, lawful and responsible processing.
Used in this study: Provides the legislative baseline, including data-protection and purpose-limitation controls.
Limitation: It does not, by itself, decide the wider ethical acceptability of a system or commercial objective.
- *OECD Principles on AI*
The focus is on fairness, human rights, transparency, robustness and accountability.
Used in this study: Characterizes international normative requirements and protections for affected stakeholders.
Limitation: Voluntary and mostly at principal level with limited operational detail.
- *NIST AI Risk Management Framework*
Main emphasis Govern Map Measure Manage
Used in this study: Provides a framework for risk ownership, assessment, testing and continuous monitoring throughout the lifespan.
Limitation: Needs to be tailored to the organization, sector and decision-making situation.
- *Algorithmic Audits Literature*
Key focus: Impact assessment, documentation, testing, monitoring and redress
Application in this study: Provides observable and assessable operational evidence for general ethical concepts.
Limitation: Effectiveness depends on sufficient authority, resources, organizational support and independent inspection.

3. Approach and Methodology

3.1 Philosophy of research and reasoning

The study is based on a pragmatic philosophy as the subject is about how governance works in practice and needs both organized comparison and contextual interpretation. Predefined control domains will deductively examine the pattern indicated in H1. If situations show unanticipated mechanisms, contradictions or contextual explanations, then abductive reasoning will be applied. This design matches the hypothesis with the analysis but is nevertheless sensitive to evidence that does not fit the basic framework.

3.2 Design of research

The unit of analysis is a documented organizational application of big-data or algorithmic analytics in a consequential business decision, such as employment, credit, pricing, fraud detection, targeting or customization. The maximum variation purposive sample will include 12–20 examples, with at least three sectors. This range is workable within the project and sufficient for analytical comparison but will not allow for population level generalization.

Research design and evidence flow

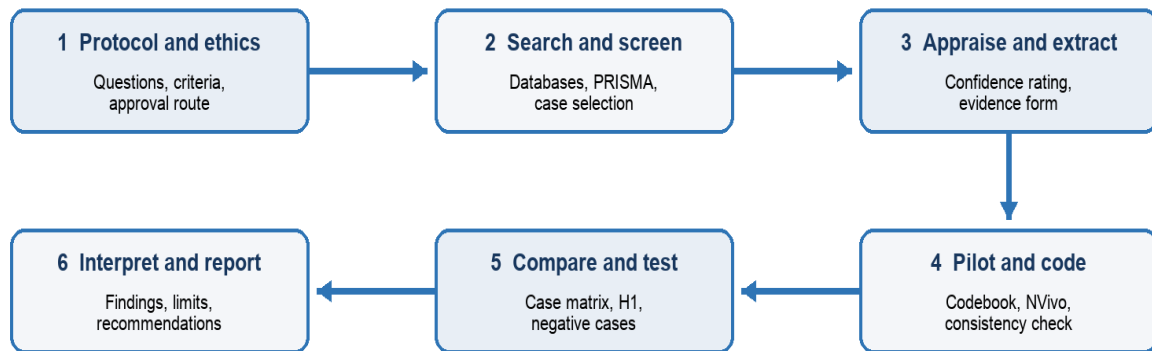


Figure 1. Sequential research-design process and evidence flow.

3.3 Search and selection procedure

The following databases will be searched: Scopus, Web of Science, Business Source Complete, ACM Digital Library, IEEE Xplore and ScienceDirect. Google Scholar will be used only for forward and backward citation searching. Where available and authoritative, grey literature will be sourced from the ICO, UK government, OECD, NIST and relevant sector regulators.

Each database will be modified with the basic search expression:

("big data" OR "data analytics" OR "algorithmic decision" OR "automated decision") AND (ethic OR privacy OR bias OR fairness OR transparent OR accountable) AND (business OR organization OR firm OR corporate) AND (governance OR audit OR "impact assessment" OR oversight)

3.4 Extraction of data and quality assessment

Studies will be assessed for design, adequacy of data, analytical transparency and constraints. Grey literature will be assessed for authority, correctness, coverage, objectivity, currency and significance. Corporate documents can say what an organization purports to do, but they cannot independently check effectiveness. As a result, each source will be given a confidence level of high, moderate or low.

The case will be recorded on a piloted extraction form including sector; decision context; affected stakeholders; data kind; reported risk; governance controls; mitigation evidence; source confidence; limitations and supporting quotation. 10% of records will be re-extracted after 2 weeks. Disagreements will be recorded and utilized to refine the codebook prior to complete analysis.

3.5 Framework and hypothesis test analysis

Analysis will follow the Framework Method; familiarization, initial coding, framework construction, indexing, charting and interpretation (Gale et al, 2013). Deductive codes will include the five governance areas and ethical hazards. Inductive subcodes will identify unforeseen practices and barriers. Each case score is tied to a quotation, page number, and confidence rating.

The initial check for H1 will be pattern matching. Cases scoring 7-10 are expected to exhibit more evidence of mitigation than those scoring 0-6. If at least 20 adequately comparable examples are identified Spearman's rank correlation will be calculated as an exploratory association. If that criterion is not attained

the study will offer medians, score distributions and qualitative case contrasts without making unjustified claims of significance. Negative instances will be active objects of study, not noise.

4. Main Technical Issues

4.1 Process of evidence handling

No software product will be built. The technical outcome is a replicable evidence and evaluation methodology. Zotero will store references, PDFs, tags and duplicate decisions. An Excel screening register will provide a unique case and document ID, record eligibility decisions and retain PRISMA totals. Files will be named using the convention `CASEID\_YEAR\_AUTHOR\_DOCTYPE` and placed in an access-controlled ARU authorized location.

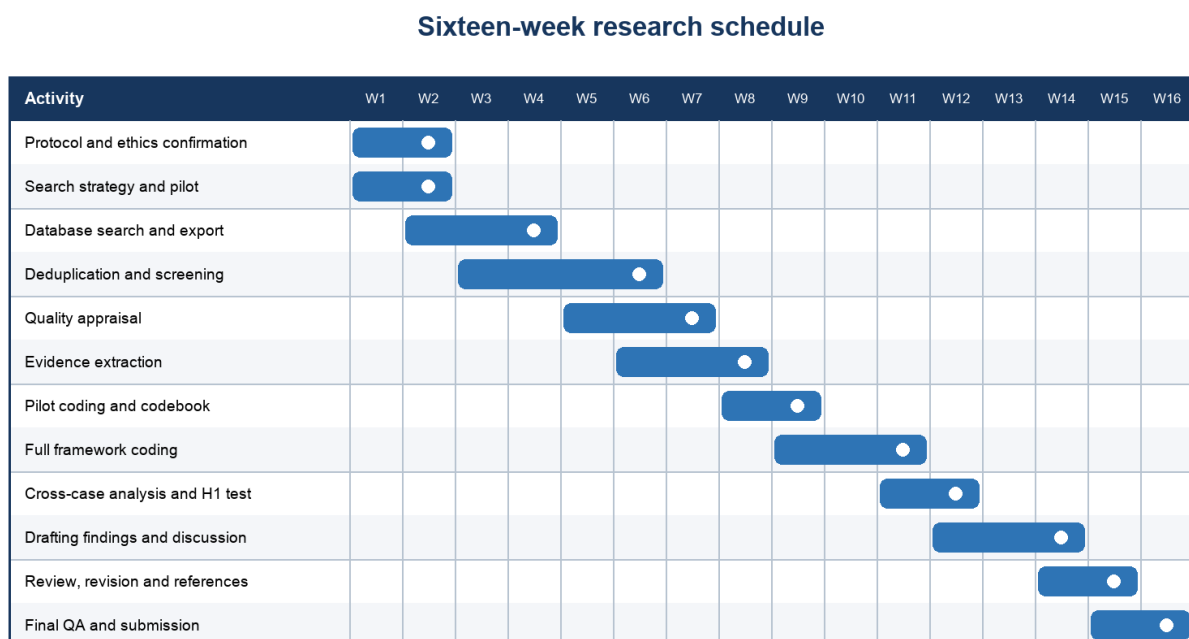
4.2 Pilot, Coding and Testing Process

The codebook will describe each 0-2 score with positive, negative and borderline instances. The search criteria, extraction fields and scoring methods will be tested using a 10% trial. The whole corpus will be coded after the modification. 10% will be re-coded after two weeks, and considerable disagreement will mean the category will be reviewed and re-coded where necessary. Excel will compute Governance Maturity Scores, distributions and if appropriate, Spearman's rho. Numeric summaries cannot replace source-level interpretation. Planned outputs include PRISMA flow diagram, evidence-quality table, case-by-control matrix, thematic risk map, hypothesis decision, limits register and practical recommendations.

5. Project Plan, Milestone and Deliverable

The project will take place over 16 weeks. The order is carefully chosen to make evidence selection and analysis independent: the search procedure will be pretested before to screening, and the corpus will be locked prior to full coding. We have minimal overlap for iteration and contingency.

Figure 2. Sixteen-week project Gantt chart



Blue bars show active work; white circles mark completion milestones.

- **Blue bars indicate active work; white circles identify completion milestones.**

Week by Week Activities

- Week 1: Finalize the study topics, hypotheses, inclusion criteria and governance-assessment framework.
- Week 2: Confirm the research methodology and ethics requirements; pilot search technique.
- Week 3: Search the chosen academic resources and regulatory sources and export records to Zotero.
- Week 4: Finalize database searching, eliminate duplicate records and record search results.
- Week 5: Screen titles and abstracts for eligibility criteria.
- Week 6: Full-text screening of available sources and quality rating of eligible sources.
- Week 7: Finalize evidence corpus and record exclusions, quality ratings, and PRISMA totals.
- Week 8: Extract evidence on organizational context, ethical hazards, governance controls and outcomes.
- Week 9: Pilot coding framework on ~10% of the evidence; Revise codebook.
- Week 10: Code the remaining scenarios utilizing the five governance domains and document supporting evidence.
- Week 11: Finish coding and develop the cross-case comparison matrix.
- Week 12: Analyze governance-maturity scores and test H1 by pattern matching and, when appropriate, determine Spearman's rank correlation.
- Week 13: Interpret positive, negative and contradictory situations; write the results and comments.
- Week 14: Complete the first complete report draft and include practical governance recommendations.
- Week 15: Edit the report, check Harvard references and ensure that all conclusions can be traced back to proof.
- Week 16: Final proofreading, formatting, checks on appendices, quality assurance, and submission.

6. Handshake and Research Ethics

The investigation will draw on publicly available academic, regulatory and organizational records. It will not include recruiting volunteers, contact vulnerable populations or collecting private individual data. A complete participant-facing review is therefore not envisaged. However, the ARU Stage 1 Research Ethics application or checklist must be submitted and confirmation acquired before research begins; minimal risk does not remove the responsibility to follow the university process (Anglia Ruskin University, n.d.). Ethical duties still apply. Access to the material will be legal, properly attributed and securely stored. Allegations made in public will not be reported as truth without official corroboration. Corporate evidence will be characterized as self-report. Predefined criteria, reflexive notes, negative-case analysis, and a decision record will mitigate selective interpretation. If the design involves interviews, private data or identifiable personal information later, work will be suspended until additional clearance.

7. Summary

This approach translates a wide-ranging topic of ethics in big data into a specific, testable inquiry of governance practice. Its design combines a falsifiable premise with transparent case selection, quality rating, organized coding and traceable evidence. This will result in a proportional assessment of which controls are linked to more demonstrated mitigation, with practical suggestions and explicit uncertainty assessment.

8. References

- I. Anglia Ruskin University. (n.d.). Research ethics approval. <https://www.aru.ac.uk/research/why-research-at-aru/research-integrity/research-ethics>
- II. Bowen, G. A. (2009) 'Document analysis as a qualitative research method', *Qualitative Research Journal*, 9(2), pp. 27-40: <https://doi.org/10.3316/ORJ0902027>.
- III. European Parliament and Council of the European Union (2024) 'Regulation (EU) 2024/1689 on harmonized rules on artificial intelligence', *Official Journal of the European Union*, L 168 <https://eur-lex.europa.eu/eli/reg/2024/1689/oj/eng>
- IV. Gale, N.K., Heath, G., Cameron, E., Rashid, S. and Redwood, S. (2013) 'Using the framework method for the analysis of qualitative data in multi-disciplinary health research', *BMC Medical Research Methodology*, 13, Art. 117. <https://doi.org/10.1186/1471-2288-13-117>
- V. Government Digital Service (2025) Data and AI Ethics Framework <https://www.gov.uk/government/publications/data-ethics-framework/data-ethics-framework-glossary-and-methodology>
- VI. Information Commissioner's Office (2023) Guidance on data protection and artificial intelligence. <https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/artificial-intelligence/guidance-on-ai-and-data-protection/>
- VII. International Organization for Standardisation (2023) ISO/IEC 42001:2023: Information technology—Artificial intelligence—Management system. Geneva: *ISO* <https://www.iso.org/standard/77919.html>
- VIII. Jobin, A., Ienca, M. and Vayena, E. (2019) 'The global landscape of AI ethical guidelines', *Nature Machine Intelligence*, 1(9), pp. 389–399. <https://www.nature.com/articles/s42256-019-0088-2>
- IX. Madaio, M.A., Stark, L., Vaughan, J.W. and Wallach, H. (2020) 'Co-designing checklists to understand organisational challenges and opportunities around fairness in AI', in *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. New York: ACM, pp. 1-14. <https://dl.acm.org/doi/10.1145/3313831.3376445>
- X. Martin, K. (2019) 'Ethical implications and accountability of algorithms', *Journal of Business Ethics* <https://link.springer.com/article/10.1007/s10551-018-3921-3>
- XI. Metcalf, J., Moss, E. and boyd, d. (2019) 'Owning ethics: corporate logics, Silicon Valley and the institutionalization of ethics', *Social Research: An International Quarterly* 86(2) <https://muse.jhu.edu/article/732185>
- XII. Mittelstadt, B. D., Allo, P., Taddeo, M., Wachter, S. and Floridi, L. (2016) 'The ethics of algorithms: mapping the debate', *Big Data & Society*, 3(2), pp. 1–21 <https://journals.sagepub.com/doi/10.1177/2053951716679679>
- XIII. Morley, J., Floridi, L., Kinsey, L. and Elhalal, A. (2020) 'From what to how: an initial review of publicly available AI ethics tools, methods and research to translate principles into practices', *Science and Engineering Ethics*, 26(4), pp. 2141–2168. <https://link.springer.com/article/10.1007/s11948-019-00165-5>
- XIV. National Institute of Standards and Technology (2023) Artificial Intelligence Risk Management Framework (AI RMF 1.0). <https://nvlpubs.nist.gov/nistpubs/ai/NIST.AI.100-1.pdf>
- XV. OECD, 2024. OECD Principles on AI. <https://www.oecd.org/en/topics/ai-principles.html>
- XVI. Page, M.J., et al. (2021) 'The PRISMA 2020 statement: an updated guideline for reporting systematic reviews', *BMJ*, 372, n71. <https://www.bmj.com/content/372/bmj.n71>

- XVII. Qiu, Y. and Hu, Z. (2025) 'Progress and recommendations in data ethics governance: a transnational analysis based on data ethics frameworks', *Humanities and Social Sciences Communications*, 12, item 1354. <https://www.nature.com/articles/s41599-025-05664-4>
- XVIII. Raji, I.D., Smart, A., White, R.N., Mitchell, M., Gebru, T., Hutchinson, B., Smith-Loud, J., Theron, D. and Barnes, P. (2020) 'Closing the AI accountability gap: defining an end-to-end framework for internal algorithmic auditing', in *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency*. New York: ACM, 33–4
<https://dl.acm.org/doi/10.1145/3351095.3372873>
- XIX. Snyder, H. (2019) 'Literature review as a research methodology: an overview and guidelines', *Journal of Business Research*, 104, pp. 333–339
<https://www.sciencedirect.com/science/article/pii/S0148296319304564?via=ihub>
- XX. Someh, I., Davern, M., Breidbach, C.F. and Shanks, G. (2019) 'Ethical concerns in big data analytics: a stakeholder view', *Communications of the Association for Information Systems*, 44, pp. 718–747. <https://aisel.aisnet.org/cais/vol44/iss1/34/>
- XXI. Zwitter, A. (2014) 'Big data ethics', *Big Data & Society*, 1(2), pp. 1–6.
<https://journals.sagepub.com/doi/10.1177/2053951714559253>

9. Appendices

1. Appendix 1 – Ethics-training certificate
2. Appendix 2 – Handshake profile evidence
3. Appendix 3 – Stage 1 research ethics checklist summary

Appendix 1 – Ethics-training certificate

Certificate supplied by the student; completed 15 April 2026.



Certificate of Completion

This certifies that

Vinay Babu Koparthi

has successfully completed the CPD Course:

Introduction to Research and Professional Ethics

Delivered by Anglia Ruskin University

Date:

15/04/2026

69df884811c08c90a80930

Appendix 2 – Handshake profile evidence

The screenshot shows the 'Explore' page on the Handshake platform for Anglia Ruskin University (ARU). The left sidebar contains navigation links: Explore, Jobs, Inbox (6), Feed, Events, People, Employers, and Career center. The main content area is titled 'Explore' and features a header for 'Anglia Ruskin University' with the tagline 'View your institution's resources'. Below this is a blue banner with the text 'Protect your account! Provide a non-institution email to ensure you can access Handshake after graduation.' and an 'Add email' button. The page displays two sections of job picks: 'Job picks for software developers and engineers in Cambridge' and 'Good fits for data scientists'. Each section includes a 'Create job alert' button. The first section lists four job opportunities: 'Graduate software engineer' at Bending Spoons, 'Graduate Formal Verification Engin...' at Axiomise Ltd, 'Graduate Technology Consultant' at Grayce, and 'Application Support Engineer (Inter...' at PLM Central Ltd. The second section lists four job opportunities: 'Full Stack Developer Intern' at Horizon Labs, 'Business Intelligence Intern' at Horizon Labs, 'Data Science Consultant intern H/F' at fifty-five, and 'Fintech Data Analyst Intern' at Premialab. At the bottom, there is a link to 'Update career interests' and a 'Get the app' button. A URL is visible at the bottom: <https://aru.joinhandshake.co.uk/stu/schools/120>.

Explore

Anglia Ruskin University
View your institution's resources

Protect your account! Provide a non-institution email to ensure you can access Handshake after graduation. [Add email](#)

Job picks for software developers and engineers in Cambridge > [Create job alert](#)

- Bending Spoons**
Internet & Software
Graduate software engineer
€66K/yr · Full-time job
London, England +3 (Remote) · 3 weeks ago
- Axiomise Ltd**
Internet & Software
Graduate Formal Verification Engin...
Full-time job
Hemel Hempstead, England (Onsite) · 1 week ago
- Grayce**
Management Consulting
Graduate Technology Consultant
£28K/yr · Full-time job
London, England (Onsite) · 1 week ago
- PLM Central Ltd**
Internet & Software
Application Support Engineer (Inter...
£32–38K/yr · Full-time job
Watford, England (Onsite) · 1 day ago · **Now**

Good fits for data scientists > [Create job alert](#)

- Horizon Labs**
Internet & Software
Full Stack Developer Intern
£300/mo · Full-time job
Remote · 3 weeks ago
- Horizon Labs**
Internet & Software
Business Intelligence Intern
£300/mo · Full-time job
Remote · 3 weeks ago
- fifty-five**
Telecommunications
Data Science Consultant intern H/F
Full-time job
Paris, Ile-de-France (Hybrid) · 3 weeks ago
- Premialab**
Internet & Software
Fintech Data Analyst Intern
Full-time job
London, England (Onsite) · 1 week ago

Not what you're looking for? [Update career interests](#)

[Get the app](#)

<https://aru.joinhandshake.co.uk/stu/schools/120>

The screenshot shows the profile page for Vinay Babu Koparthi on the Handshake platform. The left sidebar is identical to the previous screenshot. The main content area is titled 'Profile' and features a header for 'VINAY BABU KOPARTHI' with '0 followers · Following'. Below this is a section for 'ARU '27 · MSc Computer Science (with Professional Experience)' in 'Cambridge, England', with 'Share' and 'Edit' buttons. There is a section to 'Add links to profile' with a button to 'Add a link'. The profile is divided into sections: 'CV', 'Posts', and 'Profile viewers'. The 'CV' section is active and shows 'Looking for' with filters for 'Full-time job + 1 More', 'Software Developer + 1 More', and 'Cambridge, United Kingdom'. Below this is a 'View All' link. The 'About' section is empty. The 'Skills' section is empty. The 'Work experience' section is empty. The 'Education' section is empty. On the right side, there is a section titled 'People you might know' with a list of five people: HARI HARA SRI M., REEMA FIRDOUS, SRAVAN KUMAR R, SAI NIKITHA ABBU, and BHARGAVI VEGIR.. Each person has a profile picture, name, university, degree, and a 'Follow' button. A 'See More' button is at the bottom of this list.

Profile

VINAY BABU KOPARTHI
0 followers · Following

ARU '27 · MSc Computer Science (with Professional Experience)
Cambridge, England

[Share](#) [Edit](#)

Add links to profile
Make sure employers can see your full story
[Add a link](#)

CV Posts Profile viewers

Looking for

Full-time job + 1 More Software Developer + 1 More Cambridge, United Kingdom

View All

About

Skills

Work experience

Education

People you might know

- HH** HARI HARA SRI M.
ARU '26 · MSc Compu [Follow](#)
- RF** REEMA FIRDOUS
ARU '27 · MSc Compu [Follow](#)
- SK** SRAVAN KUMAR R
ARU '27 · MSc Compu [Follow](#)
- SA** SAI NIKITHA ABBU [Follow](#)
- BV** BHARGAVI VEGIR..
ARU '27 · MSc Compu [Follow](#)

[See More](#)

Appendix 3 – Stage 1 research ethics checklist summary:

Ethics Question	Response
Will the research involve human participants?	No
Will sensitive or personal data be collected?	No
Is there any risk of harm to participants?	No
Will the research use deception of any kind?	No
Does the research involve vulnerable groups?	No
Will data be stored securely and anonymously?	Yes – only published sources used
Does the research raise dual-use or misuse concerns?	Yes – critically addressed in ethics section
Does the research require formal ethics committee approval?	No – low-risk secondary research
Will all sources be properly cited (Harvard)?	Yes
Has AI ethics training been completed?	Yes – certificate appended